

LESSON 5-4 • SESSION 1 • PRACTICE • FORM B**Area of Regular Polygons**

Learning target – I can find the area of a regular polygon by finding the area of one triangle and multiplying. • 6.G.A.1

REMEMBER

- A **regular polygon** has equal sides and equal angles. Lines from the center make **congruent triangles**.
- Each triangle's base is one side. Its height is the **apothem**.
- **Area = $n \times (\frac{1}{2} \times \text{side} \times \text{apothem})$** . Use $\approx$ when the apothem is rounded.

WORKED EXAMPLE

Regular pentagon: sides 10 cm, apothem about 7 cm.

Step 1 5 sides make 5 triangles.**Step 2** One triangle: $\frac{1}{2} \times 10 \times 7 = 35 \text{ cm}^2$.**Step 3** Five triangles: $5 \times 35 = 175$.**Answer** Area $\approx 175 \text{ cm}^2$ **WORDS** **regular polygon** (*polígono regular*) • **apothem** (*apotema*) • **congruent** (*congruente*)**PRACTICE**

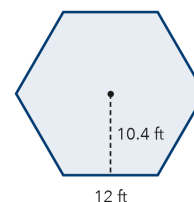
show your work • label units

1. Find the area of one center triangle:
- $\frac{1}{2} \times \text{side} \times \text{apothem}$
- .

a. side 12 cm, apothem 8 cm ____ cm^2 **b.** side 14 in, apothem 9 in ____ in^2 **c.** side 5 m, apothem 6 m ____ m^2 **d.** side 9 ft, apothem 10 ft ____ ft^2

- 2.
- Gazebo Floor.**
- A gazebo floor is a regular hexagon with sides of 12 ft.

The apothem is about 10.4 ft.

a. How many center triangles? _____**b.** Area of one triangle _____ ft^2 **c.** Area of the floor _____ ft^2 

- 3.
- Pentagon Sign.**
- A regular pentagon road sign has sides of 14 in and an apothem of about 9.6 in. Find its area.

Hint A pentagon has 5 center triangles.

Show your work.

Answer _____ in^2

- 4.
- Stepping Stone.**
- A stepping stone is a regular octagon with sides of 6 cm and an apothem of about 7.2 cm. Find its area.

Hint Find one triangle, then multiply by the number of sides.

Show your work.

Answer _____ cm^2

- 5.
- Flower Bed.**
- A flower bed is a regular hexagon with sides of 20 ft. Each center triangle has a height of about 17.3 ft. Find the area of the flower bed.

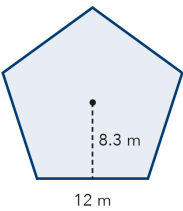
Show your work.

Answer _____ ft^2

6. **Pentagon Patio.** A patio is a regular pentagon. Decompose it into center triangles and find its area.

Hint All the triangles share the same base and height.

Show your work.



Answer _____ m²

7. **Keisha's Tile.** A tile is a regular hexagon with sides of 8 cm and an apothem of about 7 cm. Keisha says the tile's area is 28 cm². Is Keisha right? Explain.

Hint How many triangles make up the whole tile?

Show your work.

True area _____

8. **Tile Design.** A floor uses regular hexagon tiles with sides of 6 cm and an apothem of 5.2 cm.

- a. Find the area of one tile. _____ cm²
- b. Find the area covered by 30 tiles. _____ cm²

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Mateo's Work.** A regular octagon has sides of 10 in and an apothem of 12 in. Mateo wrote: "A = 10 × 12 × 8 = 960 in²." Explain the mistake and find the correct area.

TEST PRACTICE

choose the best answer

10. A regular octagon splits into how many congruent center triangles?

- A** 6 **B** 7 **C** 8 **D** 10

11. To find a regular polygon's area, multiply one triangle's area by the –

- A** apothem **B** number of sides **C** side length **D** perimeter

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

28 Q	1,038 A	336 E	374.4 H	172.8 X	672 J	168 O	249 G	2,808 N	498 Z
#2c		#3	#4	#5	#6	#7	#8b		

CHECK YOURSELF

- ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____